Grade 10 – Term 1
Decision-Making Under Uncertainty
Exam Solutions – Version 2

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Learning objective

Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.

Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

Question	C8	C9
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

Evaluated criteria

C8 Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

C9 Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

C10 Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

1 Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Essential (A_1)	100	90	70
Shared (A_2)	140	150	40
Extensive (A_3)	240	48	-40
Modular (A_4)	170	130	-20

Decision statistics (Regret = maximum regret):

Alt.	Max	Min	Regret	EMV
A_1	100	70	140	88
A_2	150	40	100	126
A_3	240	-40	110	68.8
A_4	170	-20	90	108

(a) Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A subsidy s (integer, $0 \leq s \leq 100$) increases the low-demand payoffs of A_3 and A_4 by s. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of s. **[0.5]**

(c) Among eligible camera rental plans, find the smallest allowed subsidy s at which the Maximin recommendation changes from A_1 to A_4 alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \leq s \leq 100$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

(a) Initial recommendations [1]

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	A_3	Row maximum: 240
Maximin	A_1	Row minimum: 70
Minimax Regret	A_4	Maximum regret: 90
EMV/Bayes	A_2	EMV: 126

All deciding values are in thousand USD. Maximax pursues the greatest upside; Maximin protects the worst outcome; Minimax Regret limits missed profit; EMV uses the forecast probabilities to maximize expected profit. Regret benchmarks use all four original alternatives, with state bests $(240, 150, 70)$; no eligibility restriction applies in (a).



(b) Subsidy and eligibility **[0.5]**

The updated payoff table is:

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Essential (A_1)	100	90	70
Shared (A_2)	140	150	40
Extensive (A_3)	240	48	$-40+s$
Modular (A_4)	170	130	$-20+s$

The updated decision statistics are:

Alt.	Max	Min	Regret	EMV
A_1	100	70	140	88
A_2	150	40	100	126
A_3	240	$\min(48, -40+s)$	$\max(102, 110-s)$	$68.8+0.2s$
A_4	170	$-20+s$	$\max(70, 90-s)$	$108+0.2s$

An alternative is eligible only if all its payoffs are nonnegative. For integer subsidies $0 \le s \le 100$:

Alternative	Eligibility condition	Eligible subsidies
A_1	$70 \ge 0$	$0 \le s \le 100$
A_2	$40 \ge 0$	$0 \le s \le 100$
A_3	$-40+s \ge 0$	$40 \le s \le 100$
A_4	$-20+s \ge 0$	$20 \le s \le 100$

Thus, the eligible sets are:

$$\begin{aligned} 0 \le s \le 19: &\quad \{A_1, A_2\}, \\ 20 \le s \le 39: &\quad \{A_4, A_1, A_2\}, \\ 40 \le s \le 100: &\quad \{A_3, A_4, A_1, A_2\}. \end{aligned}$$



(c) Maximin change under subsidy s **[1.5]**

The minimum payoffs are

$$m_1 = 70, \quad m_2 = 40, \quad m_3(s) = \min(48, -40 + s), \quad m_4(s) = -20 + s.$$

Since $m_3(s) \leq 48 < 70 = m_1$ and $m_2 = 40 < 70$ throughout the allowed range, only A_4 can challenge A_1.

For A_4 to beat A_1:

$$-20 + s > 70 \quad \Longleftrightarrow \quad s > 90.$$

Thus, A_1 is the sole Maximin choice for $0 \leq s \leq 89$; at $s = 90$, A_4 and A_1 tie with minimum payoff 70; and A_4 is the sole Maximin choice for $91 \leq s \leq 100$. The tie at $s = 90$ first changes the set of recommended plans; a strict change to A_4 alone requires $s > 90$. Therefore, the smallest allowed integer subsidy giving a strict change is

$$\boxed{s = 91}.$$

At $s = 91$, all four alternatives are eligible and their minimum payoffs are $70, 40, 48, 71$, respectively. Hence A_4 wins uniquely with a guaranteed profit of 71 thousand USD.



2 Outdoor Cinema Refreshments **[C8-3]**

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

Alternative	High	Low
$P(S_j)$	0.4	0.6
Portable (C_1)	96	64
Full-service (C_2)	156	20
Deluxe (C_3)	180	-10
Fixed (C_4)	140	50

Decision statistics (Regret = maximum regret):

Alternative	Max	Min	Regret	EMV
C_1	96	64	84	76.8
C_2	156	20	44	74.4
C_3	180	-10	74	66
C_4	140	50	40	86

(a) Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

(b) A sponsor replaces C_1's large-audience payoff of 96 with g (integer, $96 \leq g \leq 160$). An alternative is eligible only if all its payoffs are at least 40. Determine each refreshment kiosk's eligibility based on the value of g. **[0.5]**

(c) Among eligible refreshment kiosks, find the smallest allowed sponsored payoff g at which the Minimax Regret recommendation changes from C_4 to C_1 alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $96 \leq g \leq 160$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

(a) Initial recommendations **[1]**

Criterion	Recommended alternative(s)	Deciding statistic and value
Maximax	C_3	Row maximum: 180
Maximin	C_1	Row minimum: 64
Minimax Regret	C_4	Maximum regret: 40
EMV/Bayes	C_4	EMV: 86

All values are in hundred USD. Regret benchmarks use all four original kiosks, with state bests $(180, 64)$; no eligibility restriction applies in (a). Only EMV uses the probabilities; the other criteria compare upside, downside, or missed profit.



(b) Sponsored payoff and eligibility **[0.5]**

The updated payoff table is:

Alternative	High	Low
$P(S_j)$	0.4	0.6
Portable (C_1)	g	64
Full-service (C_2)	156	20
Deluxe (C_3)	180	-10
Fixed (C_4)	140	50

The updated decision statistics, before applying eligibility restrictions, are:

Alt.	Max	Min	Regret	EMV
C_1	g	64	$180 - g$	$0.4g + 38.4$
C_2	156	20	44	74.4
C_3	180	-10	74	66
C_4	140	50	40	86

An alternative is eligible only if both payoffs are at least 40. For integers $96 \leq g \leq 160$:

Alternative	Eligibility condition	Eligible values
C_1	$g \geq 40,\ 64 \geq 40$	$96 \leq g \leq 160$
C_2	$20 < 40$	None
C_3	$-10 < 40$	None
C_4	$140 \geq 40,\ 50 \geq 40$	$96 \leq g \leq 160$

The eligible set is always $\{C_1, C_4\}$.

(c) Minimax Regret change under sponsored payoff g [1.5]

Recalculate regrets using only these alternatives, with state benchmarks

$$b_H = \max\{g, 140\}, \qquad b_L = 64.$$

Their maximum regrets are

$$R(C_1) = \max\{140 - g, 0\},$$
$$R(C_4) = \max\{g - 140, 14\}.$$

For $96 \leq g \leq 140$, these become

$$R(C_1) = 140 - g, \qquad R(C_4) = 14.$$

For C_1 to be the sole Minimax Regret choice:

$$140 - g < 14 \quad \Longleftrightarrow \quad g > 126.$$

At $g = 126$, both alternatives have maximum regret 14. Thus, C_4 is the sole Minimax Regret choice for $96 \leq g \leq 125$; at $g = 126$, the recommendation becomes the tie $\{C_1, C_4\}$; and C_1 is the sole choice for $127 \leq g \leq 160$. The tie first changes the set of recommended kiosks, but a strict change to C_1 alone requires $g > 126$. Therefore, the smallest allowed integer giving a strict change is

$$g = 127 .$$

At this value, the eligible payoff table is:

Alternative	High	Low
C_1	127	64
C_4	140	50

The corresponding maximum regrets are

$$R(C_1) = 13, \qquad R(C_4) = 14.$$

Thus, C_1 wins uniquely with $g = 127$, corresponding to \$12,700 in large-audience profit.

3 Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan D_1–D_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
D_1	140	60	60	118
D_2	160	40	40	132
D_3	120	80	80	106
D_4	180	20	60	132
D_5	100	60	100	86
D_6	200	0	80	132

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

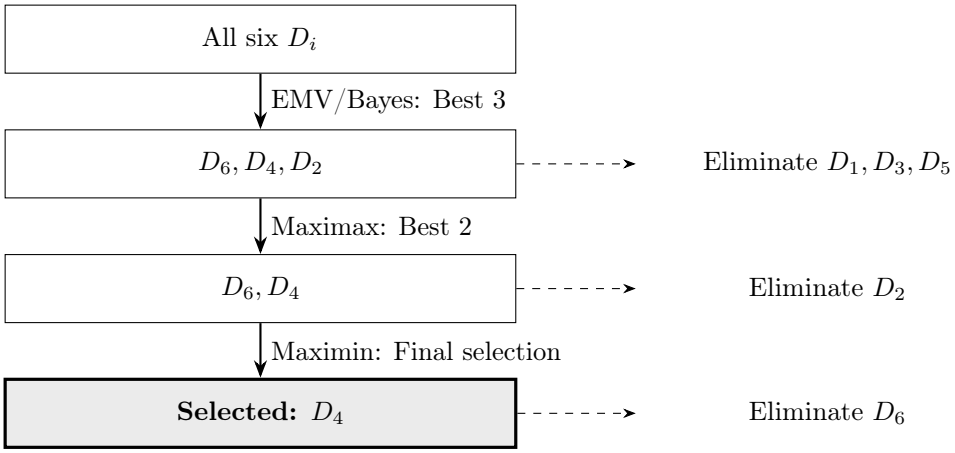
Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$D_6, D_4, D_2, D_1, D_3, D_5$	—
1	EMV/Bayes: Best 3	$D_6(132)$, $D_4(132)$, $D_2(132)$	$D_1(118)$, $D_3(106)$, $D_5(86)$
2	Maximax: Best 2	$D_6(200)$, $D_4(180)$	$D_2(160)$
3	Maximin: Final selection	**Selected:** $D_4(20)$	$D_6(0)$

The highest EMV, 132, is shared by D_6, D_4, D_2; all three fill the Best 3 places. Maximax retains D_6, D_4 with maxima 200 and 180, eliminating D_2 with 160. Maximin selects D_4 because $20 > 0$. There are no cutoff or final ties beyond the shared first-stage EMV.



Filtering tree and interpretation



The garden centre selects D_4: it survives the EMV and Maximax filters, then its minimum profit of 20 hundred USD exceeds D_6's 0.

4 Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan F_1–F_6. Net profits are in thousand USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
F_1	190	50	50	137
F_2	150	90	90	128
F_3	170	70	70	146
F_4	240	0	90	122
F_5	130	80	110	110
F_6	210	30	60	129

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

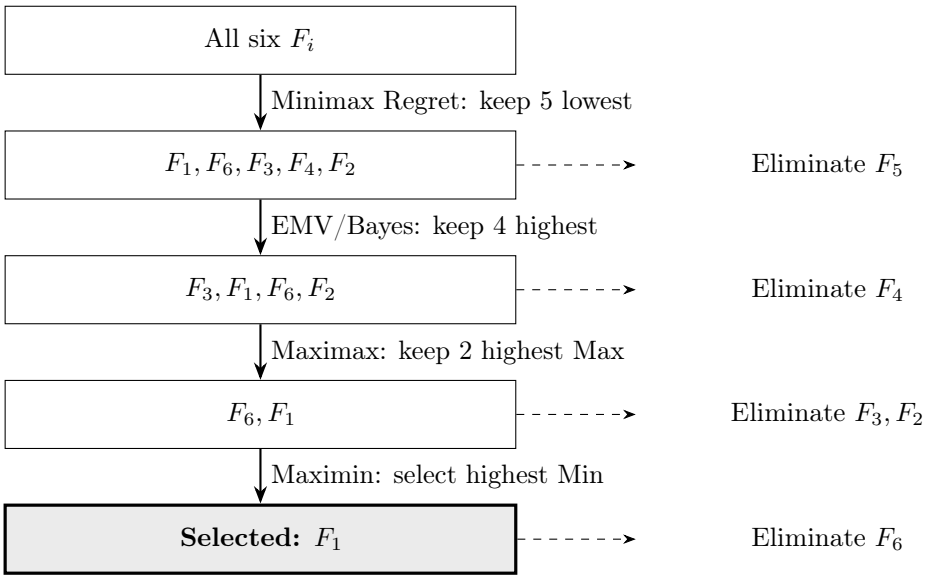
Illustrate each filtering stage, showing survivors and eliminations.

Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$F_4, F_6, F_1, F_3, F_2, F_5$	—
1	Minimax Regret: keep 5 lowest	$F_1(50)$, $F_6(60)$, $F_3(70)$, $F_4(90)$, $F_2(90)$	$F_5(110)$
2	EMV/Bayes: keep 4 highest	$F_3(146)$, $F_1(137)$, $F_6(129)$, $F_2(128)$	$F_4(122)$
3	Maximax: keep 2 highest Max	$F_6(210)$, $F_1(190)$	$F_3(170)$, $F_2(150)$
4	Maximin: select highest Min	**Selected:** $F_1(50)$	$F_6(30)$

At the first cutoff, F_4 and F_2 both have regret 90, and both survive. Later cutoffs have no ties.

Filtering tree and interpretation



The climbing gym selects F_1: after the regret, EMV, and upside filters, its minimum profit of 50 thousand USD exceeds F_6's 30.

5 Art Materials Orders [C9-2]

An art materials distributor selects an order plan H_1–H_6. Net profits are in hundred USD. The decision statistics are summarized below:

Alternative	Max	Min	Regret	EMV
H_1	150	90	90	128
H_2	170	70	70	144
H_3	210	30	60	129
H_4	130	80	110	110
H_5	240	0	90	122
H_6	190	50	50	137

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

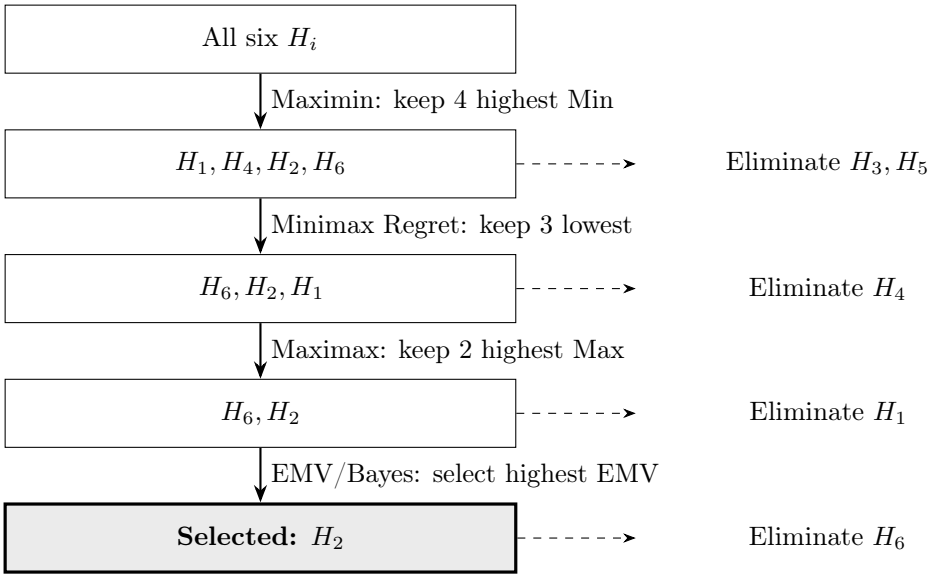
Supporting filtering table

Step	Rule	Survivors (relevant values)	Eliminated (values)
0	Initial alternatives	$H_5, H_3, H_6, H_2, H_1, H_4$	—
1	Maximin: keep 4 highest Min	$H_1(90)$, $H_4(80)$, $H_2(70)$, $H_6(50)$	$H_3(30)$, $H_5(0)$
2	Minimax Regret: keep 3 lowest	$H_6(50)$, $H_2(70)$, $H_1(90)$	$H_4(110)$
3	Maximax: keep 2 highest Max	$H_6(190)$, $H_2(170)$	$H_1(150)$
4	EMV/Bayes: select highest EMV	**Selected:** $H_2(144)$	$H_6(137)$

The cutoffs are Min 50, maximum regret 90, and Max 170; the final EMV comparison is 144 versus 137. There are no cutoff or final ties.



Filtering tree and interpretation



The art materials distributor selects H_2: after the downside, regret, and upside filters, its EMV of 144 hundred USD exceeds H_6's 137.

C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.

C10 example solution.